

PipTaz-AFT, Piperacillin and Tazobactam Powder for Solution for Injection or Infusion 4.5g

1. NAME OF THE MEDICINE

Piperacillin 4 g (as sodium salt)

Tazobactam 0.5 g (as sodium salt)

2. QUALITATIVE AND QUANTITATIVE COMPOSITION

PipTaz-AFT 4 g/0.5 g is an injectable antibacterial combination, consisting of the semisynthetic antibiotic piperacillin sodium and the β -lactamase inhibitor tazobactam sodium, for intravenous administration.

Each vial contains piperacillin sodium equivalent to piperacillin 4 g and tazobactam sodium equivalent to tazobactam 0.5 g.

Each vial of PipTaz-AFT 4 g/0.5 g contains approximately 54 mg of sodium per gram of piperacillin.

For the full list of excipients, see section 6.1.

3. PHARMACEUTICAL FORM

PipTaz-AFT 4 g/0.5 g is available as a white to off-white sterile, lyophilized powder of piperacillin and tazobactam as the sodium salts packaged in glass vials.

After reconstitution or dilution: It is a clear colourless (or not more intensely coloured than reference solution Y₅ or YG₅) solution.

4. CLINICAL PARTICULARS

4.1 Therapeutic indications

Piperacillin/tazobactam is indicated for the treatment of the following systemic and/or local bacterial infections in which susceptible organisms have been detected or are suspected.

Lower respiratory tract infections; urinary tract infections (complicated and uncomplicated); intra-abdominal infections; skin and skin structure infections; bacterial septicemia.

Polymicrobial infections: piperacillin/tazobactam is indicated for polymicrobial infections including those where aerobic and anaerobic organisms are suspected (intra-abdominal, skin and skin structure, lower respiratory tract).

Piperacillin/tazobactam, in combination with an aminoglycoside, is indicated for bacterial infections in neutropenic adults or children.

4.2 Dose and method of administration

Piperacillin/tazobactam must be given by slow intravenous infusion (e.g., over 20-30 minutes) or slow intravenous injection (over at least 3-5 minutes).

Duration of therapy

The duration of therapy should be guided by the severity of the infection and the patient's clinical and bacteriological progress.

Dosage

Neutropenic patients with signs of infection (e.g., fever) should receive immediate empirical antibiotic therapy before laboratory results are available.

Adults and Children Aged 12 Years and Older:

In general, the recommended total daily dosage is 12 g of piperacillin/1.5 g of tazobactam given in divided doses every 8 hours. The total daily dose depends on the severity and localization of the infection and can vary from 2.25 g to 4.5 g piperacillin/tazobactam administered every 6 or 8 hours.

For bacterial infections in neutropenic patients, the recommended dose is 4 g piperacillin/0.5 g tazobactam administered every 6 hours in combination with an aminoglycoside.

Pediatric Neutropenia:

<50 kg: For children with normal renal function and weighing less than 50 kg, the dose should be adjusted to 80 mg of piperacillin/10 mg of tazobactam per kilogram of body weight every 6 hours, in combination with the appropriate dose of an aminoglycoside.

>50 kg: For children weighing over 50 kg, follow the adult dosing, in combination with the appropriate dose of an aminoglycoside.

Children under the Age of 12 Years:

The following table summarises the treatment frequency and the dose per body weight for children under 12 years of age by indication or condition:

Dose per weight and treatment frequency	Indication/ condition
80 mg Piperacillin/10 mg Tazobactam per kg body weight/every 6 hours (in combination with an aminoglycoside)	Neutropenic children with fever suspected to be due to bacterial infections*

*Not to exceed the maximum 4 g/0.5 g per dose over 30 minutes.

Until further experience is available, piperacillin/tazobactam should not be used in children who do not have neutropenia.

Elderly

No dose adjustment is required for the elderly with normal renal function or creatinine clearance values above 40 mL/min.

Use in patients with renal impairment

The intravenous dose should be adjusted to the degree of actual renal impairment as follows (each patient must be monitored closely for signs of substance toxicity; medicinal product dose and interval should be adjusted accordingly):

Adults and Children 12 Years and Older:

Creatinine clearance (mL/min)	Piperacillin/tazobactam (recommended dose)
> 40	No dose adjustment necessary
20-40	Maximum dose suggested: 4 g/0.5 g every 8 hours
<20	Maximum dose suggested: 4 g/0.5 g every 12 hours

For patients on hemodialysis, one additional dose of piperacillin/tazobactam 2 g/0.25 g should be administered following each dialysis period, because haemodialysis removes 30%-50% of piperacillin in 4 hours.

Children under the Age of 12 Years:

Creatinine clearance (mL/min)	Piperacillin/tazobactam (recommended dose)
> 50	No dose adjustment needed
≤ 50	70 mg piperacillin/8.75 mg tazobactam/kg every 8 hours

For children on haemodialysis, one additional dose of 40 mg piperacillin/5 mg tazobactam/kg should be administered following each dialysis period.

The pharmacokinetics of piperacillin/tazobactam have not been studied in pediatric patients with renal impairment. Each patient must be monitored closely for signs of drug toxicity. Drug dose and interval dose should be adjusted accordingly. In patients with renal insufficiency or hemodialysis patients, intravenous dosages and administration intervals should be adjusted to the degree of renal function impairment.

Use in Children Aged Below 2 Years

The safety and efficacy of piperacillin/tazobactam in children 0 - 2 years of age has not been established. No data from controlled clinical studies are available.

Use in Patients with Hepatic Impairment

No dosage adjustment is necessary in patients with hepatic impairment.

Co-administration of Piperacillin/Tazobactam with Aminoglycosides

Due to the in vitro inactivation of the aminoglycoside by β -lactam antibiotics, piperacillin/tazobactam and the aminoglycoside are recommended for separate administration.

Piperacillin/tazobactam and the aminoglycoside should be reconstituted and diluted separately when concomitant therapy with aminoglycosides is indicated (see Section 6.2 Incompatibilities).

Geriatric Population

Patients over 65 years are not at an increased risk of developing adverse effects solely because of age. However, dosage should be adjusted in the presence of renal insufficiency.

Method of administration

For intravenous use only.

Reconstitution directions

Swirl until dissolved. When swirled constantly, reconstitution should occur within 10 minutes. Diluents for reconstitution:

- Sterile water for injections
- 0.9% Sodium chloride injection
- Glucose 5% injection

Reconstitute each vial with the volume of diluent shown in the table below, using one of the above diluents.

Vial Size (piperacillin/tazobactam)	Minimum volume of diluent to be added to vial
4.50 g (4 g/0.5 g)	20 mL

The reconstituted solution may be further diluted to the desired volume (e.g. 50 mL to 150 mL) with one of the compatible solvents for intravenous use listed below.

- Sterile water for injections*
- Saline
- 5% Glucose
- Dextran 6% in Saline

*Maximum recommended volume of sterile water for injections per dose is 50 mL.

4.3 Contraindications

The use of PipTaz-AFT 4 g/0.5 g is contraindicated in patients with a history of allergic reactions to any of the penicillins and/or cephalosporins or β -lactamase inhibitors or any of its excipients.

4.4 Special warnings and precautions for use

Serious and occasionally fatal hypersensitivity (anaphylactic/anaphylactoid [including shock]) reactions have been reported in patients on penicillin/cephalosporin therapy, including piperacillin/tazobactam. Although anaphylaxis is more frequent following parenteral therapy, it has occurred in patients on oral penicillins/cephalosporins. These reactions are more likely to occur in individuals with a history of penicillin hypersensitivity and/or a history of sensitivity to multiple

allergens. There have been reports of individuals with a history of penicillin/cephalosporin hypersensitivity who have experienced severe reactions when treated with either a penicillin or cephalosporin. Past history of a severe allergic reaction to penicillin/cephalosporin is a contraindication to the use of PipTaz-AFT 4 g/0.5 g. Before initiating therapy with any penicillin/cephalosporin, careful inquiry should be made concerning previous hypersensitivity reactions to penicillins, cephalosporins or other allergens. If an allergic reaction occurs, PipTaz-AFT 4 g/0.5 g should be discontinued and the appropriate therapy instituted. Serious anaphylactic/anaphylactoid reactions (including shock) require immediate emergency treatment with adrenaline. Oxygen, intravenous steroids and airway management, including intubation, should also be administered as indicated.

Severe cutaneous adverse reactions (SCAR), such as Stevens-Johnson syndrome (SJS), toxic epidermal necrolysis (TEN), drug reaction with eosinophilia and systemic symptoms (DRESS), and acute generalised exanthematous pustulosis (AGEP) have been reported in patients taking beta-lactam antibiotics. When SCAR is suspected, PipTaz-AFT should be discontinued immediately and an alternative treatment should be considered.

Antibiotic-associated pseudomembranous colitis has been reported with many antibiotics including piperacillin. A toxin produced by *Clostridium difficile* appears to be the primary cause. The severity of the colitis may range from mild to life threatening. It is important to consider this diagnosis in patients who develop diarrhoea or colitis in association with antibiotic use (this may occur up to several weeks after cessation of antibiotic therapy). Mild cases usually respond to drug discontinuation alone. However, in moderate to severe cases appropriate therapy with a suitable oral antibacterial agent effective against *C. difficile* should be considered. Fluids, electrolytes and protein replacement should be provided when indicated. Drugs that delay peristalsis e.g. opiates and diphenoxylate with atropine (Lomotil) may prolong and/or worsen the condition and should not be used.

Leucopenia and neutropenia may occur, especially during prolonged therapy. Therefore, periodic assessment of haematopoietic function should be performed.

As with treatment with other penicillins, neurological complications in the form of convulsions (seizures) may occur when high doses are administered, especially in patients with impaired renal function (see Section 4.8- Adverse effects (undesirable effects)).

As with other antibiotic preparations, use of this drug may result in overgrowth of non- susceptible organisms, including fungi. Patients should be carefully monitored during therapy. If superinfection occurs, appropriate measures should be taken.

Haemophagocytic lymphohistiocytosis (HLH): Rare cases of haemophagocytic lymphohistiocytosis (HLH) have been observed following therapy (>10 days) with piperacillin, often as a complication of DRESS. HLH is a pathologic immune activation which leads to excessive systemic inflammation and can be life-threatening and early diagnosis and rapid initiation of immunosuppressive therapy is essential. Characteristic signs and symptoms include fever, hepatosplenomegaly, cytopenias,

hyperferritinaemia, hypertriglyceridaemia, hypofibrinogenaemia, and haemophagocytosis. If piperacillin is suspected as possible trigger, treatment should be discontinued.

Serious and occasionally fatal hypersensitivity reactions (including anaphylactoid and severe cutaneous adverse reactions) have been reported in patients receiving therapy with beta-lactams. Before initiating therapy with PipTaz-AFT, careful inquiry should be made concerning previous hypersensitivity reactions to penicillins, cephalosporins, carbapenems or other beta-lactam agents. If an allergic reaction occurs, PipTaz-AFT must be discontinued immediately and appropriate alternative therapy instituted.

Use in renal impairment

Due to its potential nephrotoxicity (see Section 4.8 - Adverse effects (undesirable effects)), piperacillin/tazobactam should be used with care in patients with renal impairment or dialysis patients (haemodialysis and CAPD). The intravenous dose and administration interval should be adjusted to the degree of renal function impairment. Measurement of serum levels of piperacillin will provide guidance for adjusting dosage (see Section 4.2 - Dose and method of administration).

In a secondary analysis using data from a large multicenter, randomised-controlled trial when glomerular filtration rate (GFR) was examined after administration of frequently used antibiotics in critically ill patients, the use of piperacillin/tazobactam was associated with a lower rate of reversible GFR improvement compared with the other antibiotics. This secondary analysis concluded that piperacillin/tazobactam was a cause of delayed renal recovery in these patients.

Combined use of piperacillin/tazobactam and vancomycin may be associated with an increased incidence of acute kidney injury (see Section 4.5 - Interactions with other medicines and other forms of interactions).

Use with caution in the following circumstances

Bleeding manifestations have occurred in some patients receiving piperacillin. These reactions have sometimes been associated with abnormalities of coagulation tests such as clotting time, platelet aggregation and prothrombin time and are more likely to occur in patients with renal failure. If bleeding manifestations occur, the antibiotic should be discontinued and appropriate therapy instituted.

The possibility of the emergence of resistant organisms that might cause superinfections should be kept in mind, particularly during prolonged treatment. If this occurs, appropriate measures should be taken.

As with other penicillins, patients may experience neuromuscular excitability or convulsions if higher than recommended doses are given intravenously.

Repeated use of lignocaine as diluent should be avoided in patients with severe liver disease or decreased hepatic blood flow due to the possibility of lignocaine toxicity (resulting from decreased metabolism and accumulation).

Combined administration of β -lactamase inhibitors and β -lactam antibiotics may be associated with a slightly increased risk of hepatic adverse reactions. The incidence of increased liver enzymes in patients treated with PipTaz-AFT 4 g/0.5 g was slightly higher than has been reported previously with the use of piperacillin alone. The potential for increased hepatic adverse reactions should be borne in mind when using PipTaz-AFT 4 g/0.5 g.

Combined use of piperacillin/tazobactam and vancomycin may be associated with an increased risk of acute kidney injury.

Check the following before use

Periodical assessment of organ system functions including renal, hepatic and haematopoietic during prolonged therapy (>21 days) is advisable.

This product contains 54 mg of sodium per gram of piperacillin, which may increase a patient's overall sodium intake. The theoretical sodium content of each vial of PipTaz-AFT 4 g/0.5 g is 216 mg sodium. Periodical electrolyte determinations should be made in patients with low potassium reserves and the possibility of hypokalaemia should be kept in mind with patients who have potentially low potassium reserves and who are receiving cytotoxic therapy or diuretics.

Because of its poor penetration into the CSF, piperacillin is not advised in the treatment of meningitis and brain abscess.

Antimicrobials used in high doses for short periods to treat gonorrhoea may mask or delay symptoms of incubating syphilis. Therefore, prior to treatment, patients with gonorrhoea should also be evaluated for syphilis. Specimens for darkfield examination should be obtained from patients with any suspected primary lesion and serological tests should be made for a minimum of 4 months.

Use in the elderly

No data available.

Paediatric use

Safety and efficacy of the use of piperacillin/tazobactam injection in children under the age of 2 years has not yet been established.

Effects on laboratory tests

As with other penicillins, the administration of piperacillin/tazobactam may result in a false- positive reaction for glucose in the urine using a copper-reduction method. It is recommended that glucose tests based on enzymatic glucose oxidase reactions be used.

There have been reports of positive test results using Bio-Rad Laboratories Platelia Aspergillus EIA test in patients receiving piperacillin/tazobactam injection, who were subsequently found to be free of Aspergillus infection. Cross-reactions with non-Aspergillus polysaccharides and polyfuranoses with Bio-Rad Laboratories Platelia Aspergillus EIA test have been reported. Therefore, positive test results

in patients receiving PipTaz-AFT 4 g/0.5 g should be interpreted cautiously and confirmed by other diagnostic methods.

4.5 Interaction with other medicines and other forms of interaction

Aminoglycosides

The mixing of beta-lactam antibiotics with aminoglycosides in-vitro can result in substantial inactivation of the aminoglycoside.

The inactivation of aminoglycosides in the presence of penicillin class drugs has been recognized. It has been postulated that penicillin-aminoglycoside complexes form; these complexes are microbiologically inactive and of unknown toxicity.

Concurrent administration of piperacillin and tobramycin in patients with severe renal dysfunction (i.e. chronic haemodialysis patients) has been reported to reduce the elimination half-life and significantly increase the total body clearance of tobramycin.

The alteration of tobramycin pharmacokinetics in patients with mild to moderate renal dysfunction who are taking piperacillin concomitantly is unknown. However, reports suggest that the aminoglycoside inactivation in patients concomitantly taking an aminoglycoside with a broad-spectrum beta-lactam penicillin is only clinically significant in patients with severe renal dysfunction.

Probenecid

Concurrent administration of probenecid and piperacillin and tazobactam produces a longer half-life and lower renal clearance for both piperacillin and tazobactam. However, peak plasma concentrations of neither drug are affected.

Vancomycin

Studies have detected an increased incidence of acute kidney injury in patients concomitantly administered piperacillin/tazobactam and vancomycin as compared to vancomycin alone (see Section 4.4 - Special warnings and precautions for use). Some of these studies have reported that the interaction is vancomycin dose-dependent. Expert guidelines recommend intensive vancomycin dosing and maintenance of trough levels between 15 mg/L and 20 mg/L which is an increase from previously published recommendations of target trough concentrations of 5-10 mg/L. Attaining these trough concentrations often requires practitioners to prescribe vancomycin doses which exceed manufacturers' recommendations. Therefore, it is possible that in addition to the increased risk of vancomycin-induced nephrotoxicity reported with adherence to these guidelines the risk of nephrotoxicity may also increase due to an interaction with piperacillin/tazobactam.

Non-depolarizing muscle relaxants

Piperacillin when used concomitantly with vecuronium has been implicated in the prolongation of the neuromuscular blockade of vecuronium. PipTaz-AFT 4 g/0.5 g (piperacillin/tazobactam) could produce the same phenomenon if given along with vecuronium. Due to their similar mechanism of action, it is expected that the neuromuscular blockade produced by any of the non-depolarising muscle relaxants could be prolonged in the presence of piperacillin.

Methotrexate

Piperacillin may reduce the excretion of methotrexate; therefore, serum levels of methotrexate should be monitored in patients to avoid drug toxicity.

Anticoagulants

During simultaneous administration of heparin, oral anticoagulants and other medicines that may affect the blood coagulation system including the thrombocyte function, appropriate coagulation tests should be performed more frequently and monitored regularly.

4.6 Fertility, pregnancy and lactationEffects on fertility

Piperacillin and tazobactam did not affect the fertility of male or female rats.

Use in pregnancy – Category B1

Adequate human studies on the use of piperacillin/tazobactam injection during pregnancy are not available. Limited studies with piperacillin alone in rats and mice revealed no teratogenic effects or harm to the foetus. Studies with tazobactam (doses up to 3000 mg/kg IV) or tazobactam and piperacillin (doses up to 750 mg/kg and 3000 mg/kg IV) in mice showed no evidence of teratogenicity or harm to the foetus. Studies in rats at these dose levels showed no evidence of teratogenicity though maternal toxicity, in the form of decreased weight gain, was noted at the dose levels tested. Piperacillin and tazobactam cross the placenta in humans. Pregnant women should be treated only if the expected benefit outweighs the possible risks to the pregnant woman and foetus.

Use in lactation

Adequate clinical studies on the use of piperacillin/tazobactam injection during lactation are not available. Piperacillin is excreted in low concentrations in human milk; tazobactam concentrations in human milk have not been studied. In animal studies, both piperacillin and tazobactam were excreted in the milk of lactating rats. Women who are breast-feeding should be treated only if the expected benefit outweighs the possible risks to the woman and child.

4.7 Effects on ability to drive and use machines

The effects of this medicine on a person's ability to drive and use machines were not assessed as part of its registration.

4.8 Adverse effects (Undesirable effects)

Piperacillin/Tazobactam injection is generally well tolerated. The overall incidence of adverse events was 15.7% although a cause/effect relationship was not established in all cases. This incidence was comparable to that observed with other agents used in the clinical studies. Treatment had to be discontinued in only 2.9% of cases due to adverse reactions.

The most frequently reported adverse clinical reactions were diarrhoea, rash, erythema, pruritus, vomiting, allergic reactions, nausea, urticaria, superinfection, phlebitis, thrombophlebitis, dyspepsia, and insomnia.

The following adverse reactions have been reported in clinical trials and are listed in CIOMS frequency categories as follows:

- Very common $\geq 10\%$
- Common $\geq 1\%$ and $<10\%$
- Uncommon $\geq 0.1\%$ and $<1\%$
- Rare $\geq 0.01\%$ and $<0.1\%$
- Very rare $< 0.01\%$
- Unknown Cannot be estimated from available data

Skin and subcutaneous tissue disorders

Common:	Rash
Uncommon:	Pruritus, urticaria
Rare:	Eruption (including bullous dermatitis), purpura
Unknown:	Increased sweating, eczema, exanthema.

Infections and infestations

Rare:	Pseudomembranous colitis
-------	--------------------------

Gastrointestinal

Common:	Diarrhoea, including soft/loose stools, nausea, vomiting
Uncommon:	Constipation, dyspepsia, stomatitis
Rare:	Abdominal pain

Psychiatric disorders

Uncommon:	Insomnia
-----------	----------

Nervous system disorders

Uncommon:	Headache
Unknown:	Hallucination, dizziness, dry mouth

Musculoskeletal, connective tissue and bone disorders

Rare:	Arthralgia.
Unknown:	Muscular weakness, muscle pain, prolonged muscle relaxation

Vascular disorders

Uncommon:	Phlebitis, hypotension, thrombophlebitis
Rare:	Flushing
Unknown:	Tachycardia, including supraventricular and ventricular; bradycardia; arrhythmia, including atrial fibrillation, ventricular fibrillation, cardiac arrest, cardiac failure, circulatory failure, myocardial infarction

Respiratory, thoracic and mediastinal disorders

Rare:	Epistaxis
-------	-----------

Blood and lymphatic system

Uncommon:	Leucopenia, neutropenia, thrombocytopenia
Rare:	Anaemia, eosinophilia
Very rare:	Disturbed thrombocyte function

Renal and urinary disorders

Rare:	Tubulointerstitial nephritis, renal failure
-------	---

Metabolism and nutrition disorders

Very rare:	Hypokalaemia
------------	--------------

Hypokalaemia was reported in patients with liver disease and those receiving cytotoxic therapy or diuretics when given high doses of piperacillin

General disorders and administration site conditions

Uncommon:	Pyrexia, injection site reaction (pain, inflammation)
Rare:	Chills
Unknown:	Hot flushes, oedema, tiredness

Investigations

Uncommon:	Alanine aminotransferase increased, aspartate aminotransferase increased, blood creatinine increased
Rare:	Bleeding time prolonged, blood bilirubin increased*, blood alkaline phosphatase increased*, gamma-glutamyltransferase increased*.

Very rare: Coombs direct test positive, activated partial thromboplastin time prolonged, prothrombin time time prolonged, blood albumin decreased, blood glucose decreased, blood total protein decreased, blood urea increased

* The incidence is higher than with piperacillin alone.

Post-Marketing Experience

Additional adverse events reported from worldwide marketing experience with piperacillin/tazobactam, occurring under circumstances where causal relationship with piperacillin/tazobactam is uncertain.

Blood and lymphatic system

Rare: Haemolytic anaemia
Very rare: Agranulocytosis, pancytopenia, thrombocytosis

Immune system disorders

Uncommon: Hypersensitivity reaction
Rare: Anaphylactoid shock, anaphylactic shock, anaphylactoid reaction, anaphylactic reaction

Psychiatric disorders

Not known: Delirium

Nervous system disorders

Uncommon: Seizure

Infections and infestations

Uncommon: Candidal superinfection, especially with prolonged treatment

Respiratory, thoracic and mediastinal disorders

Unknown: Eosinophilic pneumonia

Renal and urinary disorders

Rare: Interstitial nephritis, renal failure
Unknown: Acute renal injury

Skin and subcutaneous tissue disorders

Uncommon: Maculopapular rash
Rare: Erythema multiforme
Very rare: Stevens-Johnson syndrome (SJS), toxic epidermal necrolysis (TEN)
Unknown: Drug reaction with eosinophilia and systemic symptoms (DRESS), acute generalized exanthematous pustulosis (AGEP), dermatitis exfoliative

Hepatobiliary

Uncommon: Jaundice

Rare: Hepatitis

Piperacillin therapy has been associated with an increased incidence of fever and rash in cystic fibrosis patients.

4.9 Overdose

There have been post-marketing reports of overdose with piperacillin/tazobactam. The majority of those events experienced including nausea, vomiting, and diarrhoea have also been reported with the usual recommended dosages. Patients may experience neuromuscular excitability or convulsions if higher than recommended doses are given intravenously (particularly in the presence of renal failure).

No specific antidote is known. Treatment should be supportive and symptomatic according to the patient's clinical presentation. In the event of an emergency, all required intensive medical measures are indicated as in the case of piperacillin. In cases of motor excitability or convulsions, anticonvulsive agents (e.g. diazepam or barbiturates) may be indicated. In cases of anaphylactic reactions, the usual counter measures are to be initiated (adrenaline, antihistamines, corticosteroids and, if required, oxygen and airway management).

Excessive serum concentrations of either piperacillin or tazobactam may be reduced by haemodialysis.

5. PHARMACOLOGICAL PROPERTIES**5.1 Pharmacodynamic properties**Mechanism of action

Piperacillin, a broad spectrum, semisynthetic penicillin active against many Gram-positive and Gram-negative aerobic and anaerobic bacteria, exerts bactericidal activity by inhibition of both septum and cell wall synthesis. Tazobactam, a triazolymethyl penicillanic acid sulfone, is a potent inhibitor of many β -lactamases, including the plasmid and chromosomally mediated enzymes that commonly cause resistance to penicillins. The presence of tazobactam in the PipTaz-AFT 4 g/0.5 g formulation enhances and extends the antibiotic spectrum of piperacillin to include many β -lactamase producing bacteria normally resistant to it. Thus, PipTaz-AFT 4 g/0.5 g combines the properties of a broad-spectrum antibiotic and a β -lactamase inhibitor.

Microbiology

PipTaz-AFT 4 g/0.5 g is active against most strains of the following β -lactamase producing and non- β -lactamase producing microorganisms:

Gram-negative bacteria

Escherichia coli, *Citrobacter* spp. *Klebsiella* spp. (including *K. pneumoniae*), *Enterobacter* spp. (including *E. cloacae*), *Proteus vulgaris*, *Proteus mirabilis*, *Serratia* spp. (including *S. marcescens*), *Pseudomonas aeruginosa* and other *Pseudomonas* spp., *Neisseria gonorrhoeae*, *Neisseria meningitidis*, *Moraxella catarrhalis*, *Acinetobacter* spp., *Haemophilus influenza*.

Gram-positive bacteria

Streptococci (*S. pneumoniae*, *S. pyogenes*, *S. agalactiae*, *S. viridans*), *Enterococci* (*E. faecalis*, *E. faecium*), *Staphylococcus aureus* (not methicillin-resistant *S. aureus*), *S. epidermidis* (coagulase-negative *Staphylococci*).

Anaerobic bacteria

Bacteroides spp. including *Bacteroides fragilis* group, *Peptostreptococcus* spp., *Fusobacterium* spp., *Eubacterium* group, *Clostridia* spp., *Veillonella* spp.

Disc Susceptibility Test

Dilution or diffusion techniques – either quantitative (MIC) or breakpoint, should be used following a regularly updated, recognised and standardised method (e.g. NCCLS). Standardised susceptibility test procedures require the use of laboratory control micro-organisms to control the technical aspects of the laboratory procedures. A report of “Susceptible” indicates that the pathogen is likely to be inhibited if the antimicrobial compound in the blood reaches the concentrations usually achievable. A report of “Intermediate” indicates that the result should be considered equivocal, and if the micro-organism is not fully susceptible to alternative, clinically feasible medicines, the test should be repeated. This category implies possible clinical applicability in body sites where the medicine is physiologically concentrated or in situations where high dosage of the medicine can be used. This category also provides a buffer zone, which prevents small-uncontrolled technical factors from causing major discrepancies in interpretation. A report of “Resistant” indicates that the pathogen is not likely to be inhibited if the antimicrobial compound in the blood reaches the concentrations usually achievable; other therapy should be selected.

Note: The prevalence of resistance may vary geographically for selected species and local information on resistance is desirable, particularly when treating severe infections. This information provides guidance on micro-organisms susceptible to piperacillin/tazobactam. The following MIC 90 values were reported in 1996 for clinical isolates collected in 3 Australian states¹.

¹ Daley, D., Mulgrave, L., Munro, S., Smith, H. and Dimech, W. An evaluation of the in vitro activity of piperacillin/tazobactam. Pathology 28: 167-172, 1996.

Table 1: MIC 90 for 1,952 clinically significant isolates

Organism (number)	MIC90(mg/L)
<i>E.coli</i> (528)	2.0
<i>Klebsiella</i> spp. (180)	4.0
<i>Klebsiella</i> spp. (ESBL 44)	64.0
<i>Enterobacter</i> spp. (142)	16.0
<i>Citrobacter/Serratia</i> spp. (84)	8.0
<i>Morganella/Proteus/Providencia</i> spp. (45)	2.0
<i>Proteus mirabilis</i> spp. (104)	2.0
<i>Pseudomonas aeruginosa</i> (88)	32.0
<i>Acinetobacter calcoaceticus</i> (40)	32.0
<i>Staphylococcus aureus</i> (433)	4.0
Coagulase-negative Staphylococcal (28)	16.0
<i>Streptococcus pneumoniae</i> (45)	0.015
<i>Enterococci</i> (109)	4.0
<i>Haemophilus influenzae</i> (59)	0.094
<i>Bacteroides fragilis</i> gp (23)	4.0

The latest NCCL references are:

Methods for Dilution Antimicrobial Susceptibility Tests for Bacteria That Grow Aerobically; Approved Standard- Seventh Edition, NCCLS document M7-A5, 2006. NCCLS, Wayne, PA

For anaerobes:

Methods for Antimicrobial Susceptibility Testing of Anaerobic Bacteria; Approved Standard- Sixth Edition. NCCLS document M11-A, 2006. NCCLS, Wayne, PA.

Clinical trials

Paediatric

A study was performed to compare the safety, tolerance, and efficacy of 100 mg/kg piperacillin/12.5 mg/kg tazobactam with those of 50 mg/kg cefotaxime plus 7.5 mg/kg metronidazole administered intravenously (IV) every 8 hours for the treatment of hospitalised paediatric patients (aged 2 to 12 years of age) with clinically or bacteriologically diagnosed intra- abdominal infection (IAI). The cure rates in the efficacy evaluable (EE) population at the follow- up visit were 90% and 91% for piperacillin/tazobactam and cefotaxime plus metronidazole, respectively. The results of the clinical and microbiological analyses in 521 patients showed that piperacillin/tazobactam administered intravenously was at least as effective as cefotaxime plus metronidazole in the treatment of children aged 2 to 12 years with severe IAIs.

5.2 Pharmacokinetic properties

Absorption

Mean plasma concentrations of piperacillin and tazobactam at steady state of the combination appear in Table 2 and Table 3. Peak piperacillin and tazobactam plasma concentrations are attained

immediately after completion of an intravenous infusion. When given with tazobactam, piperacillin plasma levels are similar to those attained when equivalent doses of piperacillin are administered alone.

Table 2: Plasma levels in adults after a thirty-minute intravenous infusion of piperacillin/tazobactam (steady state)

PIPERACILLIN PLASMA LEVELS (µg/mL)						
Piperacillin/Tazobactam Dose	30*min	1 hr	1.5 hr	2 hr	3 hr	4 hr
2 g/250 mg	134	57	29	17	5	2
4 g/500 mg	298	141	87	47	16	7
TAZOBACTAM PLASMA LEVELS (µg/mL)						
Piperacillin/Tazobactam Dose	30*min	1 hr	1.5 hr	2 hr	3 hr	4 hr
2 g/250 mg	14.8	7.2	4.2	2.6	1.1	0.7
4 g/500 mg	33.8	17.3	11.7	6.8	2.8	1.3

*Completion of 30 minute infusion

Table 3: Plasma levels in adults after an intramuscular injection of piperacillin/tazobactam (steady state)

PIPERACILLIN PLASMA LEVELS (µg/mL)						
Piperacillin/Tazobactam Dose	30 min	1 hr	1.5 hr	2 hr	3 hr	4 hr
2 g/250 mg	55	45	31	19	8	4
TAZOBACTAM PLASMA LEVELS (µg/mL)						
Piperacillin/Tazobactam Dose	30 min	1 hr	1.5 hr	2 hr	3 hr	4 hr
2 g/250 mg	10.5	7.4	4.9	3.2	1.4	0.9

Distribution

In healthy subjects piperacillin/tazobactam plasma elimination half-life range from 0.7 to 1.2 hours following single or multiple doses. These half-lives are unaffected by dose or duration of infusion. Piperacillin and tazobactam are 21% and 23% respectively, bound to plasma proteins. The protein binding of either piperacillin or tazobactam is unaffected by the presence of either compound. Piperacillin and tazobactam are widely distributed in tissues and body fluids including intestinal mucosa, gall bladder, lung and bile.

Metabolism

Piperacillin does not undergo biotransformation in humans. Approximately 20% of a dose of tazobactam is metabolised to a single metabolite that has been found to be microbiologically inactive.

Excretion

Piperacillin and tazobactam are eliminated by the kidney via glomerular filtration and tubular secretion. Piperacillin is excreted rapidly as unchanged drug, with 69% of the dose appearing in the urine. Piperacillin is also secreted into bile. Tazobactam and its metabolite are eliminated primarily by renal excretion, with 80% of the dose appearing as unchanged drug and the remainder of the dose appearing as the metabolite.

Impaired renal function

The half-life of piperacillin and tazobactam increases with decreasing creatinine clearance. The increase is two-fold and four-fold for piperacillin and tazobactam, respectively, at creatinine clearance below 20 mL/min compared to patients with normal renal function. Dosage adjustments are recommended when creatinine clearance is below 40 mL/min (see Section 4.2 Dosage and method of administration).

Piperacillin and tazobactam are removed from the body during haemodialysis with 31% and 39% of the doses of piperacillin and tazobactam, respectively, recovered in the dialysis fluid. Piperacillin and tazobactam are removed from the body by peritoneal dialysis with 5% and 12% of the dose, respectively, appearing in the dialysate. For dosage recommendations in patients undergoing haemodialysis (see Section 4.2 Dosage and method of administration).

Impaired liver function

Piperacillin half-life and AUC were increased by 25% and 40% respectively and tazobactam half-life and AUC by 18% and 23% respectively in patients with hepatic impairment. However, dosage adjustments in patients with hepatic impairment are not necessary.

Children

The pharmacokinetics of piperacillin and tazobactam have been examined in 24 paediatric patients aged 2 months to 12 years receiving 100 mg/kg piperacillin/12.5 mg/kg tazobactam (Table 4). The maximum concentration (C_{max}) for both piperacillin and tazobactam is increased relative to the maximum adult dose but the predicted time above the minimum inhibitory concentration is slightly decreased. The dosage of 100 mg/kg piperacillin/12.5 mg/kg tazobactam administered every 8 hours is predicted to provide coverage 31% to 61% of the time for the range of MIC values of 2 µg/mL to 16 µg/mL commonly found in intra-abdominal infections in children.

Table 4: Piperacillin and tazobactam pharmacokinetics in children (cv%) following single doses

Dose	Patient age	C _{max} (mg/L)	AUC (mg.h/L)	CL (mL/min/kg)	V _{ss} (L/kg)	T _{1/2} (h)
Piperacillin 100 mg/kg	2-5 mo	382(15)	539(29)	3.3(24)	0.28(32)	1.3(16)
	6-23 mo	344(15)	373(27)	4.8(29)	0.25(27)	1.0(24)
	2-5 y	408(80)	331(21)	5.2(19)	0.23(36)	0.9(26)
	6-12 y	394(24)	404(17)	4.2(21)	0.24(42)	0.8(27)
Tazobactam 12.5 mg/kg	2-5 mo	43(49)	63(32)	3.6(28)	0.32(31)	1.3(15)
	6-23 mo	35(22)	42(23)	5.2(24)	0.33(29)	1.1(23)
	2-5 y	45(42)	37(24)	5.8(19)	0.27(33)	0.9(29)
	6-12 y	45(25)	57(27)	3.9(36)	0.28(36)	1.3(57)

6. PHARMACEUTICAL PARTICULARS

6.1 List of excipients

Sodium bicarbonate.

6.2 Incompatibilities

PipTaz-AFT 4 g/0.5 g should not be mixed with other drugs in a syringe or infusion bottle since compatibility has not been established. Whenever PipTaz-AFT 4 g/0.5 g is used concurrently with another antibiotic, the drugs must be administered separately.

Because of chemical instability, PipTaz-AFT 4 g/0.5 g should not be used with solutions containing only sodium bicarbonate or having a pH in the basic range.

PipTaz-AFT 4 g/0.5 g should not be added to blood products or albumin hydrolysates.

6.3 Shelf life

Observe expiry date imprinted on outer carton.

6.4 Special precautions for storage

Before reconstitution, store below 30 °C.

Piperacillin and Tazobactam for Injection is stable within 24 hours at 2-8°C and stable within 12 hours at room temperature after diluted to reconstitution concentration by specified reconstitution solution. (0.9% sodium chloride for injection, sterile water for injection, dextrose 5%).

Piperacillin and Tazobactam for Injection is stable within 12 hours at room temperature and 24 hours at 2-8°C after diluted to intravenous infusion concentration by specified solution (0.9% sodium chloride for injection, sterile water for injection, dextrose 5%, dextran 6% in saline).

To avoid microbial contamination, Piperacillin and Tazobactam for Injection should be used immediately after reconstitution. If storage is necessary, discard any unused portion after storage for 12 hours at room temperature, or after storage for 24 hours at 2-8°C. The total storage time for reconstituted and diluted solution should not exceed 12 hours at room temperature, or 24 hours at 2-8°C.

For single use in one patient only. Discard any remaining residue.

6.5 Nature and contents of container

PipTaz-AFT 4 g/0.5 g powder for injection is available as a vial containing piperacillin sodium equivalent to 4 g piperacillin and tazobactam sodium equivalent to 500 mg tazobactam. It is available in packs of 1 and 10 vials.

6.6 Special precautions for disposal

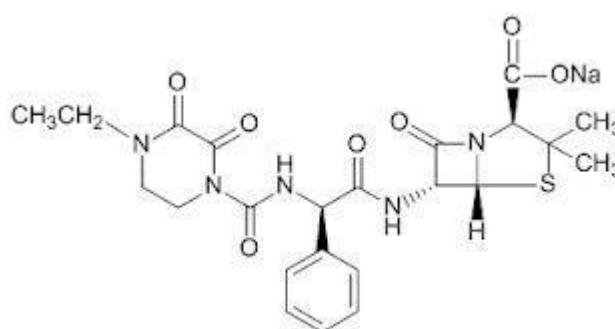
Any unused medicine or waste material should be disposed of in accordance with local requirements.

6.7 Physicochemical properties

Piperacillin sodium

Chemical structure

Piperacillin sodium has the chemical name sodium (2S,5R,6R)-6-[(R)-2-(4-ethyl-2,3-dioxo-1-piperazine-carboxamido)-2-phenylacetamido]-3,3-dimethyl-7-oxo-4-thia-1-azabicyclo[3.2.0]heptane-2-carboxylic acid.



The empirical formula is C₂₃H₂₆N₅NaO₇S.

The molecular weight is 539.54.

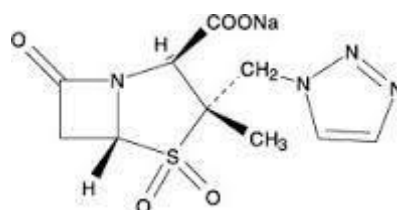
CAS number

59703-84-3

Tazobactam sodium

Chemical structure

Tazobactam sodium has the chemical name sodium (2S-(2 α ,3 β ,5 α))-3-methyl-7-oxo-3-(1H-1,2,3-triazol-1-ylmethyl)-4-thia-1-azabicyclo[3.2.0] heptane-2-carboxylic acid 4,4-dioxide.



The empirical formula is C₁₀H₁₁N₄O₅S.

The molecular weight is 322.28.

CAS number

89785-84-2

8. Product Registration Holder

AFT Pharmaceuticals (S.E. Asia) SDN BHD
Suite 9.01-Regus, Level 9, Menara Summit,
Persiaran Kewajipan, USJ 1, UEP,
47600 Subang Jaya, Selangor, Malaysia.

9. DATE OF REVISION

February 2024